Michael Liu

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Bachelor of Science in Computer Science — Expected May 2029
Concentration in Machine Learning

- **Selected Coursework:** Mathematical Foundations for Computer Science (15-151), Great Theoretical Ideas in Computer Science (15-251), Functional Programming (15-150), Matrices and Linear Transformations (21-241), Concepts of Robotics (16-180), Imperative Computation (15-122), Probability Theory (36-225).

The Stony Brook School — Stony Brook, NY
High School Diploma — June 2025

SKILLS

- **Programming Languages:** Python, C, C++, Java, SML, C0, JavaScript, Node.js, LaTeX.
- **Technologies:** PyTorch, FastAPI, React Native, Firebase, TensorFlow, Scikit-Learn, Pandas, NumPy, Git, Docker, Jupyter, MCP, RAG.
- **Spoken Languages:** Native fluency in English and Mandarin Chinese.

PROJECTS & RESEARCH

SURA: Neurodiversity Wellbeing Project — May 2026 – August 2026

- Reduced developer context-switching friction by integrating the Model Context Protocol (MCP) into Jupyter server extensions for local context awareness (Advised by Prof. Andrew Begel, VariAbility Lab / S3D).
- Maintained sub-100ms interface responsiveness by engineering custom JupyterLab UI components delivering real-time, non-intrusive emotional support tailored for neurodivergent developers.
- Validated interaction flow and system efficacy by designing and conducting structured user studies with autistic software engineers.

Viditas: Multimodal Deepfake Detection System — August 2026

- Accelerated AI-generated media classification throughput to 30+ FPS by developing an asynchronous inference backend in FastAPI and PyTorch.
- Enhanced detection precision across benchmark datasets by building end-to-end pipelines processing parallel audio and video streams for real-time anomaly detection.

ScottyTasks (React Native & Firebase) — January 2026 – Present

- Scaled task management tooling for the CMU community, integrating gamification elements to drive recurring daily active engagement.
- Decreased data sync latency to under 50ms by implementing real-time NoSQL synchronization and user authentication via Google Firebase.

Stack-Based Simple Compiler (C0) — October 2025 – November 2025

- Achieved a 100% execution pass rate across 50+ bytecode test suites by engineering a stack-based instruction set compiler in C0.
- Eliminated runtime memory corruption and assertion faults by enforcing strict dynamic function contracts and safety preconditions.

Science Olympiad: Robot Tour — 2024 – 2025

- Placed 2nd (2024) and 3rd (2025) out of 20+ teams at Regionals by programming autonomous navigation algorithms and closed-loop hardware controls.
- Calibrated trajectory-tracking logic to rank among top competing teams at the Yale Invitational 2025.

HONORS & LEADERSHIP

- **Tepper Hackathon (CMU Energy Week):** 3rd Place (March 2026) – Built quantitative model optimizing energy efficiency and business sustainability.
- **USACO Platinum Division (2024):** Perfect score in Gold Division (Top ∼300 nationally).
- **USAMO Qualifier (2024):** AMC 12 top 1% nationwide; scored 11/15 on AIME.
- **CMU ScottyLab Labrador Member** — September 2025 – Present